



A long life 4 V class lithium-ion polymer battery with liquid-free polymer electrolyte



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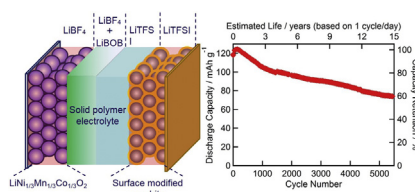
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HIGHLIGHTS

- Lithium-ion dry polymer battery with 4 V cathode and graphite anode is proposed.
- Addition of LiBOB at cathode/SPE interface is effective for the reversibility.
- Surface modification of graphite plays a role to minimize the lithium consumption.
- Proposed battery reached 5400 cycles with 60% capacity retention at 323 K.

GRAPHICAL ABSTRACT



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ABSTRACT

Ether-based solid polymer electrolyte (SPE) is one of the most well-known lithium ion conductors. Unlike the other inorganic electrolytes, SPE exhibits advantages of flexibility and large-area production, enabling low cost production of large size batteries. However, because the ether group is oxidized at 4 V versus Li/Li^+ cathode, and due to its high irreversibility with the carbon anode, ether-based SPE was believed to be inapplicable to 4 V class lithium-ion batteries with carbon anode. Here we report a remarkably stable SPE in combination with a 4 V class cathode and carbon anode achieved by the proper design at the interface. The introduced boron-based lithium salt prohibits further oxidation of SPE at the cathode interface. The surface modification of graphite by the annealing of polyvinyl chloride mostly prohibits the continuous consumption of lithium at the graphite anode. Using above interface design, we achieved 60% capacity retention after 5400 cycles. The proposed battery provides a possible approach for realizing flammable electrolyte-free lithium-ion batteries, which achieve innovative safety improvements of large format battery systems for stationary use.

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1. Introduction

Fundamental safety is an urgent issue for the full-scale introduction of large lithium-ion batteries for electric vehicles (EV) and battery energy storage (BES) for stationary use. In particular, the batteries for use in the grid require capacity on the order of MWh

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[1], making safety the first priority in the system design. Although various “safe” electrode materials such as LiFePO_4 [2] or $\text{Li}_4\text{Ti}_5\text{O}_{12}$ [3] have already been proposed, the main safety criterion of lithium-ion batteries is to prevent an ignition of the flammable electrolyte vapor. The application of lithium conductive solid electrolyte had been proposed as a fundamental solution of this problem, and maximum ionic conductivities (σ) approaching 10^{-2} Scm^{-1} [4] and 10^{-3} to 10^{-4} Scm^{-1} [5,6] have been achieved in the sulfide and oxide systems, respectively, at room temperature. Alkali ion conduction in ether-based polymer had been proposed in 1970's [7]. While the ionic conductivity is relatively low ($\sigma = 10^{-4} \text{ Scm}^{-1}$ at 333 K) [8], other characteristics, such as shape flexibility and production capability of large area thin-film electrolyte, are major advantages for the development of a large lithium-ion battery. Although inorganic solid electrolytes (ISE) have been realized in the sodium sulfur battery (NaS , $\beta\text{-Al}_2\text{O}_3$), ISE was used in the combination of liquid electrodes in the NaS system. On the other hand, electrode materials of lithium-ion batteries are solids, and they raise volume change during the intercalation/deintercalation reaction of lithium ions. Therefore, the choice of flexible polymer is absolutely advantageous for maintaining a good interface with electrode materials in a large lithium-ion battery for long operation periods.

The main issues for SPE were (i) insufficient stability at high voltage conditions with the 4 V class cathodes [9] and (ii) incompatibility with the carbon anode [10]. The reported oxidation potential of polyether-based SPE was 3.6 V [9]. Therefore, in the previous work, we have introduced an oxidation protection barrier material such as Li_3PO_4 [11], $\text{LiAl}_{0.5}\text{Ge}_{1.5}(\text{PO}_4)_3$ [12], and Al_2O_3 [13] at the LiCoO_2 surface to prevent the oxidation of SPE and obtain improved reversibility. Similar approaches have also been reported, such as the combination with the ISE system [14]. However, the coating process applied to the cathode powder will increase production cost. Therefore, we have proposed to introduce a carboxymethylcellulose (CMC) at the $\text{LiNi}_{1/3}\text{Mn}_{1/3}\text{Co}_{1/3}\text{O}_2$ (NMC) SPE interface as the sacrificial additive to prevent the oxidation of SPE, and demonstrated operation for over 1000 cycles [15]. In this article, we propose an advanced additive combination with lithium bis(oxalate)borate (LiBOB) at the interface between $\text{LiNi}_{1/3}\text{Mn}_{1/3}\text{Co}_{1/3}\text{O}_2$ (NMC) and SPE by simply coating the SPE solution with additives on the cathode side [16].

On the other hand, the reversibility of carbon anode with SPE has originally been reported by Imanishi et al. [17]. We have also reported sufficient reversibility of graphite and also the mixture of $\text{SiO}/\text{Graphite}$ with SPE [18]. In addition, we also analyzed capacity fading in $[\text{LiNi}_{1/3}\text{Mn}_{1/3}\text{Co}_{1/3}\text{O}_2|\text{SPE}|\text{graphite}]$ cell by introducing a pseudo reference electrode and found that continuous consumption of lithium at the graphite anode is the main origin of capacity fading [19]. Similar trends were also reported for the graphite anode with liquid electrolyte system even though the graphite formed a stable solid electrolyte interface (SEI) at the surface [20,21]. By contrast, SPE cannot play a sufficient role in the formation of the SEI with materials such as ethylene carbonate (EC) and/or vinyl carbonate (VC) so that the continuous lithium consumption with SPE is larger than that with a liquid electrolyte on the graphite surface. Therefore, surface layer on the graphite with action comparable to SEI is intrinsically required in the SPE system. Here, we apply a surface modification additive derived from polyvinyl chloride (PVC, Wako Chemical) on graphite. Such modification was already proposed in the liquid electrolyte system, and the reversibility of graphite with propylene carbonate (PC) was improved by the modification [22]. We applied a similar modification to graphite, compared the lithium consumption at the graphite after operation, and determined the additive content using Raman spectroscopy.

2. Experimental

2.1. Cathode electrode preparation

90.6 wt% $\text{LiNi}_{1/3}\text{Mn}_{1/3}\text{Co}_{1/3}\text{O}_2$ (NMC) was mixed with 6.4 wt% conductive additives (3.2 wt% carbon black and 3.2 wt% VGCF[®], Showa Denko) and a 3.0 wt% water-based binder (2.0 wt% carboxymethylcellulose; CMC and 1.0 wt% styrene butadiene rubber; SBR). The mixed aqueous slurry was pasted on aluminum foil and pressed to the density of 2.0 g cm^{-3} (based on the active material). The loaded cathode active material weight and thickness were 2.5 mg cm^{-2} and $13 \mu\text{m}$ respectively. After electrodes were introduced into a glove box (MIWA MFG Co., Ltd., $\text{H}_2\text{O} < 0.1 \text{ ppm}$, $\text{O}_2 < 0.4 \text{ ppm}$), 10 wt% SPE ($[\text{P}(\text{EO}/\text{MEEGE})] = 88/12$ (Osaka Soda where EO is ethylene oxide, MEEGE is 2-(2-methoxyethoxy)ethyl glycidyl ether, MW = 1 Million) [8] in acetonitrile (AN) were coated on the electrodes. Lithium tetrafluoroborate (LiBF_4 , Kanto Denka Kogyo) was mixed with SPE in a molar ratio of $[\text{O}]/[\text{Li}] = 10/1$ ($[\text{O}]$ denotes oxygen in the ethylene oxide unit) in the coated solution. 2 wt% of Tetrakis [methylene-3(3',5' -di-*tert*-butyl-4'-hydroxyphenyl)propionate] methane antioxidant agent (Antioxidant 1010) was also added into the coating solution. The agent is commonly used as an antioxidant in synthetic fibers and elastomers [23,24]. Here, LiBOB (0, 1, 3, and 5 wt% versus SPE weight) was added, and the correlation between the LiBOB additive content and battery performance was determined. Here, LiBOB content denoted on the $[\text{O}]/[\text{Li}]$ ratio was 10/0.023 (1% LiBOB), 10/0.068 (3% LiBOB), and 10/0.11 (5% LiBOB) respectively. The coated SPE was dried at 353 K for over 10 h in a vacuum.

2.2. Anode electrode preparation

We selected three types of graphite, $10 \mu\text{m}$ natural graphite (NG1), $12 \mu\text{m}$ natural graphite (NG2) and $20 \mu\text{m}$ artificial graphite (AG). As received graphite was mixed with PVC in the graphite/PVC ratios of 2/1 to 1/1.7 and annealed in argon atmosphere at 1173 K for 1 h [22]. The proposed surface modification is a simple annealing in the presence of PVC and graphite in argon atmosphere so that it is difficult to evaluate the coating characteristics from the preparation condition. Therefore, we applied Raman spectroscopy (JASCO, NRS-2100) to elucidate surface morphology, and determined the correlation between structural morphology of the modified graphite and the reversibility of the lithium ion cells. 95.1 wt% of graphite, 1.9 wt% of VGCF[®] and 2.9 wt% of water-based binder (1.9 wt% CMC and 1.0 wt% SBR) were mixed in the aqueous slurry and pasted on electrolytic copper foil and pressed to the density of 1.0 g cm^{-3} . The loaded anode active material weight and thickness were 1.3 mg cm^{-2} and $13 \mu\text{m}$ respectively. The coating AN solution for the anode consisted of SPE (MW = 1.5 Million) and lithium bis(trifluoromethylsulfonyl)imide (LiTFSI, 3M) in the $[\text{O}]/[\text{Li}]$ molar ratio of 16/1.

2.3. Cell preparation and operation

SPE sheet between electrodes consisted of $[\text{P}(\text{EO}/\text{MEEGE}/\text{AGE}) = 82/18/2$ (Osaka Soda) where AGE is allyl glycidyl ether]. LiTFSI (3M) was added to the SPE at the $[\text{O}]/[\text{Li}]$ molar ratio of 16/1 [8]. The thickness of the SPE sheet was approximately $50 \mu\text{m}$. The cell preparation procedure was described in detail previously [19]. The thickness of overcoated SPE on the electrode was approximately $10 \mu\text{m}$. We prepared $[\text{NMC}|\text{SPE}|\text{Gr}]$ and also $[\text{NMC}|\text{SPE}|\text{Li}]-[\text{Li}|\text{SPE}|\text{Gr}]$ laminate pouch cells. Here, total thickness of SPE in $[\text{NMC}|\text{SPE}|\text{Gr}]$ was approximately $70 \mu\text{m}$, and that in $[\text{NMC}|\text{SPE}|\text{Li}]$ and $[\text{Li}|\text{SPE}|\text{Gr}]$ were approximately $60 \mu\text{m}$ respectively. In the latter case, lithium electrodes were connected through the current

collector and act as a pseudo reference electrode. The active material weight ratio of graphite to NMC was approximately 1:2. The reversible capacities of graphite and NMC using a lithium counter electrode were 340 mAh g^{-1} (between 0 and 2.5 V) and 150 mAh g^{-1} (between 2.7 and 4.2 V), respectively. Therefore, the reversible capacity of graphite was designed to be approximately 13% larger than that of NMC. We named this setup the “Nico-Ichi” cell (this means that the two half cells act as a single lithium-ion cell) [19]. The Nico-Ichi-type cells and normal [NMC|SPE|Gr] cells were operated between 2.5 and 4.2 V. In the case of [NMC|SPE|Li] cells, they were operated between 2.7 and 4.3 V. The prepared cells were operated at 333 K or 323 K. At 333 K, they were cycled in a protocol of (C/8 for two cycles and C/2 for 48 cycles) $\times n$ loops. At 323 K, the cycling rate during operation was set to C/4.

2.4. Cell disassembly and quantitative analysis of Li in graphite

After 400 cycle operation, [Gr|SPE|Li] cell was electrochemically deintercalated up to 2.5 V vs. Li/Li⁺ and disassembled in the glove box. SPE sheet was gently removed from the graphite electrode, and the coated SPE was removed by rinsing in the dimethyl carbonate (DMC)/dimethoxy ethane (DME) = 1/1 solution for 6 h. Subsequently, the graphite electrode was immersed into 1 M HNO₃ to extract the lithium in the SEI, which was then quantified using atomic absorption spectroscopy (AAS).

3. Results and discussion

3.1. Effect of LiBOB addition to cathode interface

We applied an accelerated test condition with up to 4.3 V of charge in order to estimate the effect of LiBOB on high voltage reversibility. Fig. 1 shows (a) discharge capacity trends, (b) average Coulombic efficiency during cycle operations, and (c) initial discharge capacity of [NMC|SPE|Li] cells at 333 K. The plotted data were for the C/8 rate. The cycle performance and the Coulombic efficiency were drastically improved by the LiBOB addition. However, initial capacity showed a negative effect of the LiBOB addition, suggesting that addition of too much LiBOB led to increased interfacial impedance. We also determined 10 wt % LiBOB and confirmed poor initial capacity and also short cycle performance. Therefore, LiBOB content in the coated SPE was fixed to 5 wt% of coated SPE in all of the following experiments.

3.2. Effect of capacity retention by graphite modification

As mentioned above, graphite anodes with SPE cannot easily form stable SEI at the surface. This leads to a continuous consumption of lithium at the anode side, causing insufficient cycle performance. Therefore, we compare the cycle performance of non-coated (as received) graphite and surface modified graphite. The combined cathode is NMC (5 wt% LiBOB). Fig. 2 shows (a) (d) cathode, (b) (e) anode and (c) (f) cell voltage profiles of the Nico-ichi cells during discharge at C/8. The capacity retention of the non-coated graphite (72% at the 400th cycle) was markedly lower than that of the surface modified graphite (81% at the 400th cycle). In all cycles, anode potential increased rapidly at the end of discharge, indicating that the cell capacities were limited by the anode at the end of discharge [19]. On the other hand, cathode potentials with non-coated graphite at the end of discharge increased more than those for the surface modified graphite, indicating that cell capacity with non-coated graphite decreases due to the insufficient supply of lithium ions to NMC from the anode during discharge [20,21]. This suggests that lithium

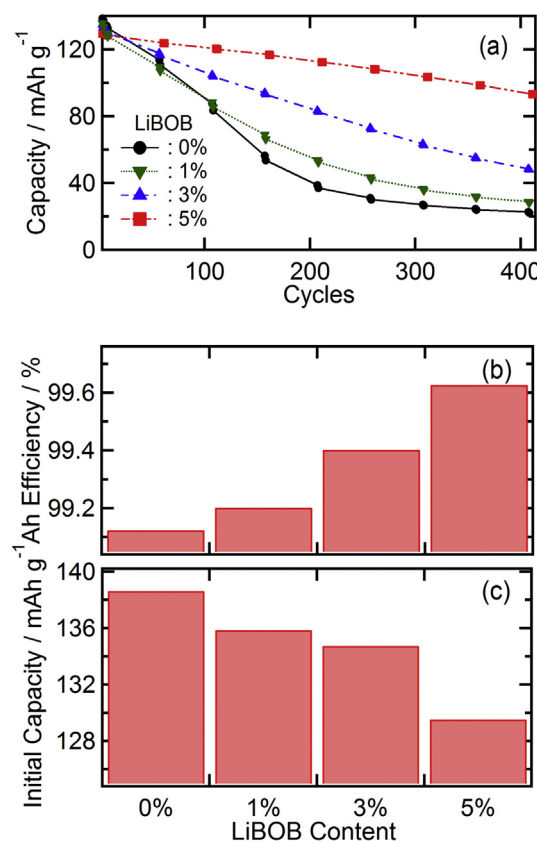


Fig. 1. Correlation between capacity retention of [NMC|SPE|Li] cell and LiBOB content in the coating SPE on the cathode. Operation condition: 4.3/2.7 V CC cutoff, (48 cycles (C/2) → 2 cycles (C/8)) $\times n$ loops. Operation temperature: 333 K.

consumption at the anode is larger in the non-coated graphite than that in the surface modified graphite.

To validate the above assumption, Nico-ichi cells after 400 cycle operation were discharged to 2.5 V and held for 24 h to maintain identical cell voltage (2.5 V) when the cell was separated as shown in Fig. 3. The open circuit potential (OCP) of NMC with non-coated graphite (3.73 V) was higher than that of the NMC with the surface modified graphite (3.68 V). The residual capacity of NMC with non-coated graphite (blue dashed dot line in Fig. 4(a)) was 43 mAh g^{-1} while that for the surface modified graphite was 25 mAh g^{-1} (blue dashed dot line in Fig. 4(b)). By contrast, the reversible capacities of NMC with different types of graphite are quite similar, with 117 mAh g^{-1} (87% retention after 400 cycles) obtained for non-coated graphite and 119 mAh g^{-1} (89% retention) obtained for the surface modified graphite as shown in Fig. 4(a) and (b) (solid lines). The capacity retentions of NMC were higher than those of the Nico-Ichi cells. Therefore, the consumption of lithium at the graphite anode during cycle operation should contribute to the capacity fading, and the lithium consumption of the non-coated graphite was larger than that of the surface modified graphite.

We then quantitatively compared the actual lithium consumptions at the graphite as follows. Lithium content after full deintercalation of graphite was estimated by AAS as shown in Fig. 4(c). The obtained content of the lithium dissolved in the acid solution (bottom axis in Fig. 4(c)) was converted to the lithium content based on the weight of the counter NMC (top axis in Fig. 4(c)). The calculated values (48 mAh g^{-1} for the non-coated graphite and 32 mAh g^{-1} for the surface modified graphite) are nearly equal to the actual residual capacity of NMC (43 mAh g^{-1} for the non-coated

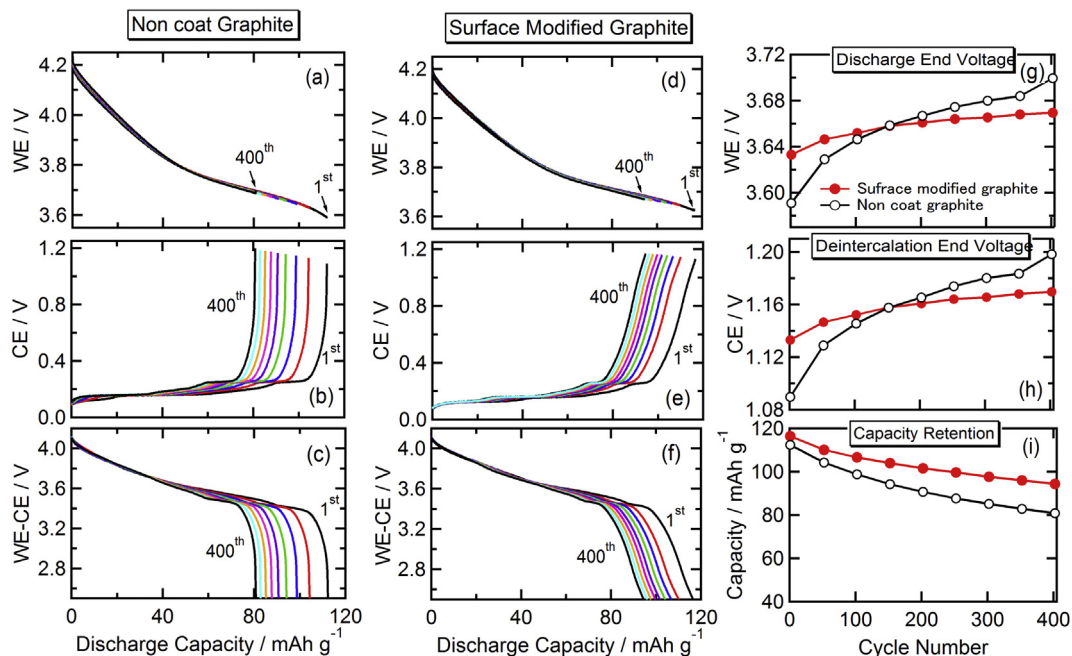


Fig. 2. Cycle performance of [NMC|SPE|Li]-[Li|SPE|Gr] (Nico-Ichi) cells using non-coated graphite (a,b,c) and surface modified graphite (e,f,g) at the anode. (a), (d): Discharge voltage profiles of cathodes. (b), (e): Deintercalation voltage profiles of anodes. (c), (f): Discharge voltage profiles of cells. (g): Trends of discharge end voltage of cathodes. (h): Trends of deintercalation end voltage of anodes. (i): Capacity retention of Nico-Ichi cells. Operation condition: 4.2/2.5 V CC cutoff, (48 cycles (C/2) → 2 cycles (C/8)) × *n* loops. Operation temperature: 333 K.

graphite and 25 mAh g⁻¹ for the surface modified graphite) [20]. Therefore, we confirmed that the consumption of lithium at the graphite can be estimated by the residual capacity of NMC after the

separation from the lithium ion cells, and the reversibility of the lithium-ion cell was improved by the suppression of lithium consumption at the anode by surface modification.

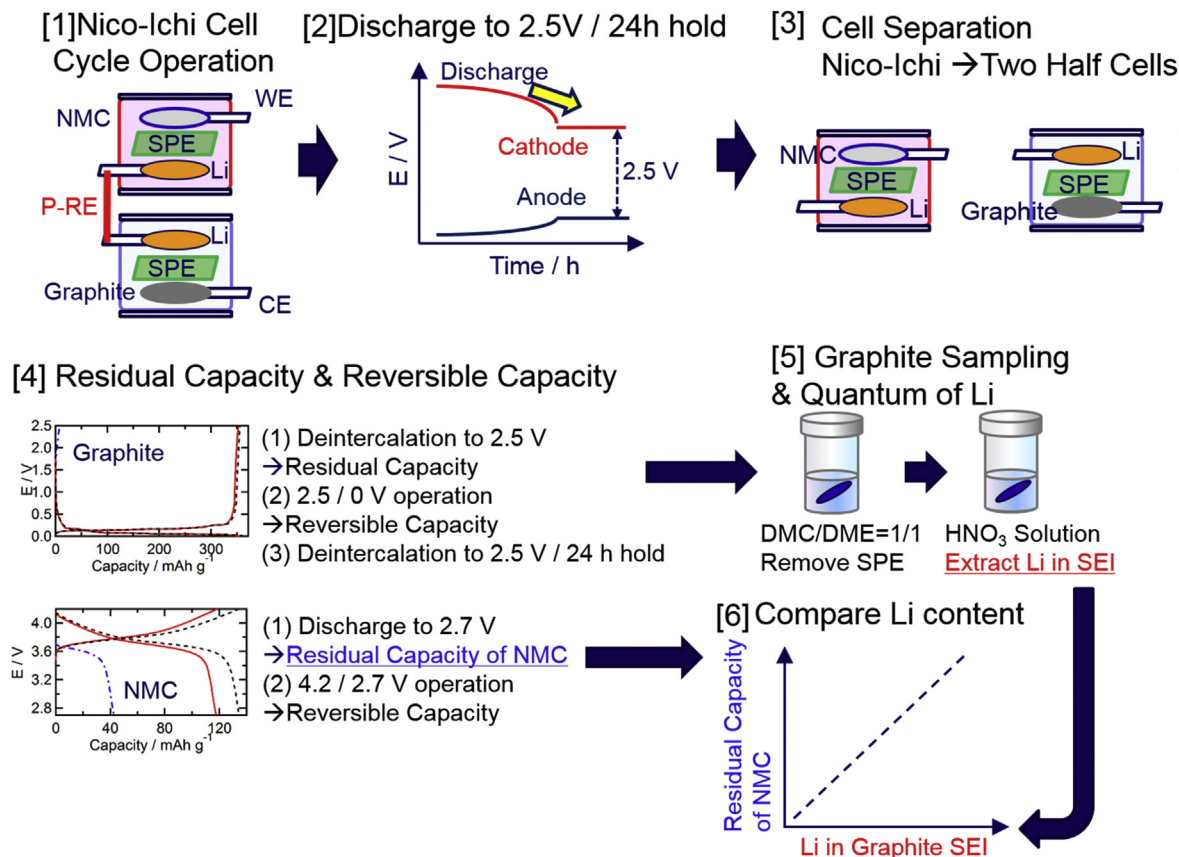


Fig. 3. Cell separation and disassembly process of Nico-Ichi cells after cycle operation.

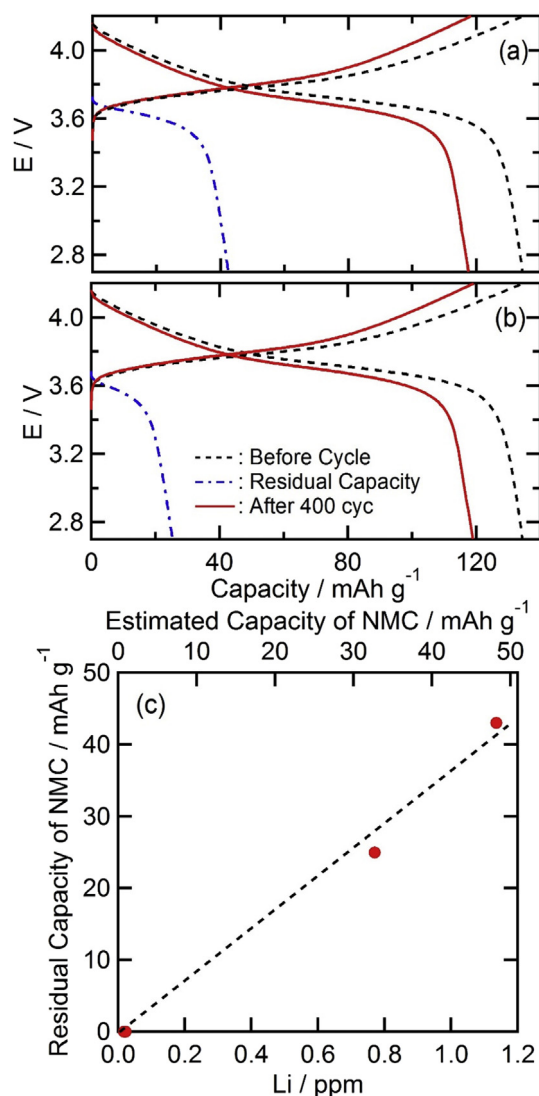


Fig. 4. Charge/discharge voltage profiles of [NMC|SPE|Li] cells with (a) non-coated graphite, (b) surface modified graphite Nico-Ichi cells, and the correlation between Li content in the graphite anode after deintercalation and residual capacity of NMC after the cells separation from Nico-Ichi cell(c). Dashed lines in (a) and (b) denote voltage profiles before cycle operation. Red lines denote voltage profiles after 400 cycle operation. Blue dashed-dotted lines denote the discharge voltage profile just after cells separation from the Nico-Ichi cell (2.5 V between NMC and graphite) after 400 cycles. Operation condition: 4.2/2.7 V CC cutoff, C/20. Operation temperature: 333 K. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.3. Correlation between surface modification of graphite and cycle performance

Typical Raman spectra of various graphite modifications are shown in Fig. 5. We assigned five bands at 1220 cm⁻¹, 1350 cm⁻¹, 1480 cm⁻¹, 1580 cm⁻¹, and 1595 cm⁻¹. Among these, the three bands at 1350 cm⁻¹, 1580 cm⁻¹ and 1595 cm⁻¹ were assigned to the sp² graphite-like structure, and the others to the sp³ type; these are often observed in amorphous carbonaceous materials [25–27]. Here, the degree of graphitization was defined using the sp² band area ratio between the D band (1350 cm⁻¹) and the G band (1580 cm⁻¹ and 1595 cm⁻¹) as D/(D + G). On the other hand, an sp²/sp³ ratio was defined using the sp² and sp³ band areas as (sp²)/(sp²+sp³). We estimated these two values for various carbon materials with the results listed in Table 1. The relationship between

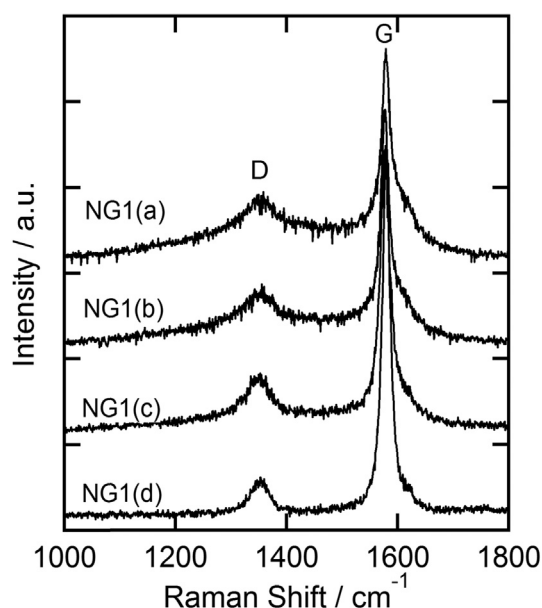


Fig. 5. Raman spectra of graphite in various PVC derived modification conditions.

the degree of graphitization and the sp²/(sp²+sp³) ratio in various carbon materials is shown in Fig. 6. Among these materials, NG1(a) and NG1(b) were categorized by a low degree of graphitization and a low sp²/(sp²+sp³) ratio. The obtained value of the sp²/(sp²+sp³) ratio is similar to that of hard carbon (Carbotron® P). On the contrary, received graphite and PVA modified graphite showed no sp³ character. This suggested that PVA easily disappeared in the conditions of an argon gas atmosphere at 1173 K, and no amorphous carbonaceous formation of the original graphite occurred in the treatment. This means that the existence of chloride in PVC is important for the formation of the amorphous carbonaceous material with sp³ structure. NG1(c) and NG2(a) in Table 1 were intermediate character between the sp³ rich group and pure graphite, and the sp²/(sp²+sp³) ratios of these materials were similar to that of carbon black (Acetylene Black).

Capacity retentions of [NMC|SPE|Graphite] using various surface modified graphite materials were compared and are shown in Fig. 7. The plotted results were compared based on C/8 capacity. The cycle capacities based on C/2 was over 70% versus C/8 capacity. However, we selected only C/8 data to avoid overabounded data. The best cycle performance was obtained for the cell using NG1(c). By contrast, cells with sp³ related carbon rich graphite (NG1(a) and NG1(b)) showed cycle performance similar to that obtained for the non-coated graphite (AG). The similar trends of AG observed also NG2(b) and NG2(c). This suggests that there exists an optimum coating ratio of sp³ related carbon on the graphite. Additionally, quantitative analysis using Raman spectroscopy is a useful tool for the evaluation of optimum surface modification of graphite with SPE.

Here, SPE used in the [NMC|SPE|Graphite] cell contained different kind of Li salts. LiTFSI was used in anode side and also separator SPE whereas LiBF₄+LiBOB additive was used in cathode side. Although most of these salts might diffuse before operation, such salt gradient design in SPE exhibited positive effect not only in the “Nico-Ichi” cell but also the normal full cell. LiTFSI was reported to react with Al current collector [28]. We believe that LiBF₄+LiBOB additive act as a corrosion inhibitor with Al even after the diffusion in SPE.

Finally, we demonstrated a long cycle operation using the optimum surface modified graphite, NG2(a), which showed a

Table 1
Degree of graphitization and $sp^2/(sp^2+sp^3)$ ratio of modified carbons estimated from Raman spectra.

Primary material (Particle size)	Mixing material	Degree of graphitization/%	$sp^2/(sp^2+sp^3)$ ratio/%	Abbreviation
NG ^a (10 μm)	PVC ^b	54.07	83.40	NG1(a)
↑	PVC	56.80	84.46	NG1(b)
↑	PVC	70.51	90.49	NG1(c)
↑	As received	87.26	100.00	NG1(d)
NG (12 μm)	PVC	71.37	89.25	NG2(a)
↑	PVA ^c	82.01	100.00	NG2(b)
↑	As received	80.03	100.00	NG2(c)
Artificial graphite	As received	87.08	100.00	AG
Hard carbon ^d	As received	31.51	88.58	HC
Carbon fiber ^e	As received	93.80	100.00	CF
Carbon black ^f	As received	51.20	91.85	CB

^a Natural graphite.

^b Polyvinyl carbonate.

^c Polyvinyl alcohol.

^d Carbotron[®] P.

^e VGCF[®].

^f Acetylene black.

Graphitization Ratio/ $sp^2/(sp^2+sp^3)$ ratio similar to that of NG1(c) as shown in Fig. 8. Here, operation temperature was changed to 323 K to restrain the side reaction of the cell. We also changed the cycle operation rate to C/4 to demonstrate the performance in the actual stationary-use operation conditions such as peak shift operation of

photovoltaic generation to demand peak in evening. We obtained over 85% capacity in the C/4 cycle operation versus C/8 capacity check operation. The operation state of charge during cycle operation was between 0 and 100%. The recorded capacity retention at 5400th cycle was 60% of initial capacity, which corresponds to an operational life of 15 years based on 1 cycle/day. Based on the 80% retention as a usable life of the battery, estimated operation life (cycle) was corresponding to 8 years (3000 cycles). Additionally, actual testing term of the cell was over 3 years. Furthermore, capacity trends showed no folding point in all operations, suggesting that the capacity fading mechanism did not change in all operation periods. This is very favorable for extrapolation of the operation life from the short term cycle operation.

4. Summary

We achieved a long life 4 V class lithium-ion battery with liquid free solid polymer electrolyte (SPE) by designing $LiNi_{1/3}Mn_{1/3}Co_{1/3}O_2$ (NMC)/SPE and graphite/SPE interfaces. The capacity retention of the [NMC/SPE/Graphite] cell was 60% at 5400 cycles, corresponding to 15 years durability based on 1 cycle/day. The obtained results strongly support the hypothesis that polyether-based SPE can be used with 4.2 V charge cutoff voltage region by introducing lithium bis(oxalate)borate (LiBOB) at the interface, and can also be used with the graphite anode through appropriate SEI formation at the surface by the surface modification of graphite. All electrodes and additives used in this work are already applied in the lithium-ion battery, and no expensive materials are contained in the

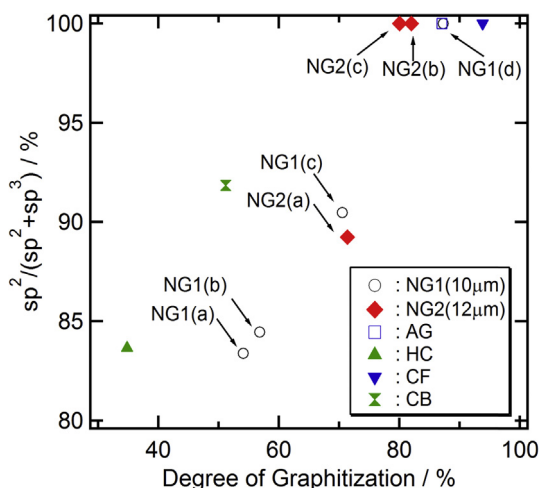


Fig. 6. Correlation between degree of graphitization degree $sp^2/(sp^2+sp^3)$ ratio of various carbons.

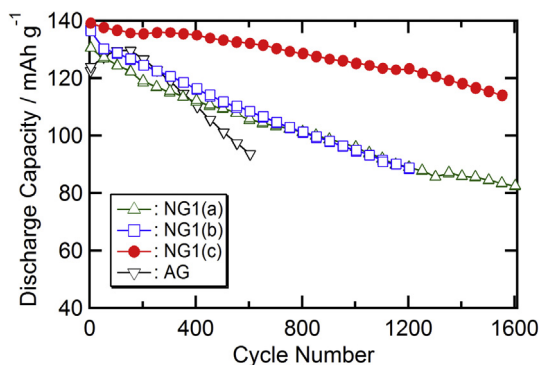


Fig. 7. Cycle performance of [NMC/SPE/Gr] cells with various kinds of surface modified graphite. Operation condition: 4.2/2.5 V CC cutoff, (48 cycles (C/2) → 2 cycles (C/8)) × *n* loops. Operation temperature: 333 K.

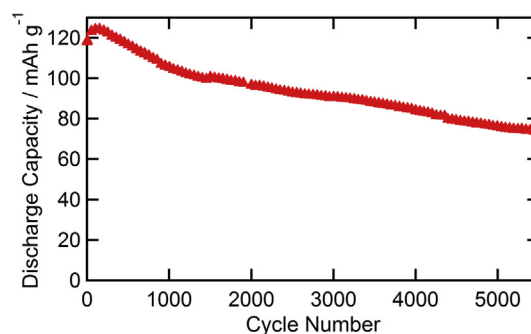


Fig. 8. Cycle performance of properly designed [NMC/SPE/Gr] cell. Graphite type: NG2(a). Operation condition: 4.2/2.5 V CC cutoff, (48 cycles (C/4) → 2 cycles (C/8)) × *n* loops. Operation temperature: 323 K.

proposed battery. Therefore, we can design an “intrinsically safe” and “long life” dry polymer battery with a cost that is competitive with that of the conventional lithium-ion batteries. We believe that the proposed robust battery system will play an important role in the stationary applications of the near future.

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